Notebook IV: Semiclassical Bulk Normalization and the N=2 Anchors

Final normalization checkpoint for the symmetric Carrollian W3 screening programme

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1. Result, scope, and immutable dependencies

This notebook proves the convention-complete semiclassical dictionary between the frozen boundary screening family and the classical bulk Noether screenings. It is a normalization and interpretation calculation. It does not reopen the tangent OPE or any Gate-3 theorem.

The load-bearing Gate-3B dependency is the portable archive

`Notebook_III_Gate3B_Definitive_Freeze_Grade_Portable.zip`
`SHA-256 a77a670b5e7ff4811cdf30094702ccdd772bdd8b9f157b4329f502bb2989a0cb`

The four earlier frozen archives and their exact hashes are:

Notebook I

`Notebook_I_Freeze_Grade_Merged_Final.zip`
`SHA-256 662de55b99f9e61f8e3c66d646972e286973f75e6a04165c5cbd4b3e172a3b5f`

Notebook II, Gate 1

`Notebook_II_Gate1_Freeze_Grade_Peer_Review_Corrected.zip`
`SHA-256 860df721f9de6c266ff4f9016aedce6c1bc908afa66ec36095ec3094de42c871`

Notebook II, Gate 2

`Notebook_II_Gate2_Closed_Outcome_B_Normalization_Corrected_Final.zip`
`SHA-256 522f0a57b5e128bd9fda861d036d30b1ee36f98f98d34cfd4cf5d98862a16481`

Notebook III, Gate 3A

`Notebook_III_Gate3A_Final_Freeze_Grade_Provenance_Corrected.zip`
`SHA-256 45e9bfdf2906527ea2175513dff447e9052d06a14c94dd979904d07323962c82`

The frozen operator relation is

$$Q_i^{[1]} = \widetilde{Q}_i^{[1]} - \mathsf{A}_{\mathrm{T},i} Q_i^{[0]}, \qquad \mathsf{A}_{\mathrm{T},i} = \mu\, \alpha_i \cdot q^- + \nu\, \alpha_i \cdot q^+.$$

Here $\widetilde{Q}_i^{[1]}$ is the raw horizontal coefficient between moving target fibres. The fixed-target operator $Q_i^{[1]}$ is its PBW covariantization and is **not an ordinary second quantum screening**.

2. Convention firewall

2.1 Circle, Fourier, and bracket conventions

The bulk circle has period 2π. For an ordinary bulk field,

$$X(\phi) = \sum_{n\in\mathbb{Z}} X_n e^{in\phi}, \qquad X_n = \frac{1}{2\pi}\int_0^{2\pi} e^{-in\phi} X(\phi)\, d\phi.$$

We use

$$[A,B]_{\mathrm{cl}} = i\{A,B\}, \qquad \delta'(\phi-\phi') = \partial_\phi \delta(\phi-\phi').$$

Consequently,

$$\{\psi'(\phi), \bar{\psi}(\phi')\} = 8\pi G \delta(\phi-\phi')$$

implies

$$\{\psi'(\phi), \bar{\psi}'(\phi')\} = -8\pi G \delta'(\phi-\phi'), \qquad [\psi'_m, \bar{\psi}'_n]_{\mathrm{cl}} = 4Gm\, \delta_{m+n,0}.$$

The sign follows from differentiating in the second argument and then integrating by parts. It is not imported from a mode formula.

2.2 FH real current versus the frozen Euclidean cylinder current

Fredenhagen and Hörl use the real cylinder expansion [FH, eqs. (3.2), (3.11)]

$$J_{\mathrm{FH}}(\phi) = i \sum_{n\in\mathbb{Z}} J_n e^{in\phi}.$$

The frozen plane/cylinder bridge from Notebook II is

$$w=\tau-i\phi,\qquad z=e^{w},\qquad \mathcal{J}_w=zj_{\rm pl}-\frac{\alpha}{2},\qquad J_{\rm FH}=i\mathcal{J}_w.$$

Thus

$$\mathcal{J}_{w,n}=a_n-\frac{\alpha}{2}\delta_{n0}.$$

The factor i in the FH field expansion is indispensable. If it is omitted while the ordinary bulk Fourier convention is retained, the crossed mode bracket changes sign.

2.3 Ordering and derivative namespaces

The quantum minus currents use the symmetric averaged product of [FH, eqs. (2.14), (4.29)-(4.31)]. The reduction

$$\langle NP\rangle_{\rm av}\longrightarrow NP$$

is made only after the strict classical limit. The verifier distinguishes four derivative namespaces,

$$D_{\rm boundary},\quad D_{\rm bulk},\quad D_{\rm circle},\quad D_{\rm plane},$$

and four ordering tags: commutative, averaged, parent-normal, and parent-antinormal. A jet from one namespace cannot be differentiated by another.

3. Semiclassical normalization triangle

Write the bulk Chern-Simons level and invariant-trace normalization as [DDF, eqs. (4.23), (6.3)-(6.5)]

$$k_{\rm CS}=\frac{1}{8\pi G},\qquad \tau_s=\frac{12}{s(s^2-1)},\qquad a_s=\sqrt{\pi k_{\rm CS}\tau_s}.$$

Then

$$a_s^2=\frac{3}{2Gs(s^2-1)}.$$

For the symmetric Carrollian free fields [FH, eqs. (3.30)-(3.31), (4.16)-(4.17)],

$$c_L=-4s(s^2-1)\alpha^+\alpha^-,\qquad c_M=-2s(s^2-1)(\alpha^+)^2.$$

The gravity branch is

$$\boxed{\alpha^-=0,\qquad \alpha^+=+ia_s}.$$

It gives

$$c_L=0,\qquad c_M=-2s(s^2-1)(ia_s)^2=\frac{3}{G}.$$

The semiclassical parameter is

$$g_s=a_s^{-1}=O(\sqrt{G}),\qquad \mu=(\alpha^+)^{-1}=-\frac{i}{a_s}=O(\sqrt{G}).$$

```
import sympy as s
G,r=s.symbols('G r',positive=True)
a2=s.Rational(3,2)/(G*r*(r**2-1))
assert s.simplify(-2*r*(r**2-1)*(s.I**2*a2)-3/G)==0
assert s.simplify(a2.subs(r,2)-1/(4*G))==0
assert s.simplify(a2.subs(r,3)-1/(16*G))==0
print('CELL_IV_1_PASS')
```

3.1 Resolution of the printed N=2 square-root tension

At $s = 2$,

$$a_2 = \frac{1}{2\sqrt{G}}, \qquad \alpha^+ = \frac{i}{2\sqrt{G}}.$$

The sentence after [FH, eq. (2.16)] prints $i/\sqrt{2G}$. The two displayed equations immediately preceding that sentence give

$$c_M = -12(\alpha^+)^2, \qquad c_M = \frac{3}{G}.$$

They require $(\alpha^+)^2 = -1/(4G)$, not $-1/(2G)$. The printed value produces $6/G$. The general trace normalization, the complete Miura coefficient solve, and the crossed bracket below all give $i/(2\sqrt{G})$. No scalar-to-weight rescaling modifies the central-charge identity. The special sentence is therefore inconsistent with the source's own equations.

4. Internal bulk N=2 reduction

Choose the $\mathfrak{sl}_2$ weights

$$\epsilon_1 = \frac{1}{\sqrt{2}}, \qquad \epsilon_2 = -\frac{1}{\sqrt{2}}.$$

Equation [DDF, (6.49)] reads

$$(D_0)_{jj} = \frac{i}{a_2}\epsilon_j\psi', \qquad (D_1)_{jj} = \frac{i}{a_2}\epsilon_j\bar{\psi}'.$$

Matching $D_0 = \psi' J_0$, $D_1 = \bar{\psi}' P_0$, with $J_0 = \mathrm{diag}(1/2, -1/2)$, fixes the same conversion in both sectors:

$$\boxed{\psi = -\frac{ia_2}{\sqrt{2}}\psi, \qquad \bar{\psi} = -\frac{ia_2}{\sqrt{2}}\bar{\psi}.}$$

The unbarred and one-bar row determinants become

$$(a_2\partial + i\epsilon_1\psi')(a_2\partial + i\epsilon_2\psi') = a_2^2\left(\partial^2 + M_{\mathrm{bulk}}\right),$$

$$\sum_{j=1}^{2}(a_2\partial + i\epsilon_1\psi')\cdots(i\epsilon_j\bar{\psi}')\cdots(a_2\partial + i\epsilon_2\psi') = a_2^2 L_{\mathrm{bulk}},$$

where [DDF, eq. (4.41)]

$$M_{\mathrm{bulk}} = -\frac{1}{4}(\psi')^2 - \frac{1}{2}\psi'', \qquad L_{\mathrm{bulk}} = -\frac{1}{2}\psi'\bar{\psi}' - \frac{1}{2}\bar{\psi}''.$$

```
import sympy as s
a=s.symbols('a',nonzero=True)
eps=(1/s.sqrt(2),-1/s.sqrt(2)); C=-s.I*a/s.sqrt(2)
assert [s.simplify(s.I*e*C/a) for e in eps]==[s.Rational(1,2),-s.Rational(1,2)]
x,xp,y,yp=s.symbols('x xp y yp')
assert s.expand((x/2)*(-x/2)-xp/2+x**2/4+xp/2)==0
assert s.expand((y/2)*(-x/2)-yp/2+(x/2)*(-y/2)+x*y/2+yp/2)==0
print('CELL_IV_2_PASS')
```

5. Boundary/bulk N=2 normalization firewall

Start without imposing a current normalization:

$$J^+ = A\psi', \qquad J^- = B\bar{\psi}' + C\psi', \qquad \alpha^+ = Da_2,$$

$$M_{\text{bdry}} = \rho_M M_{\text{bulk}}, \qquad L_{\text{bdry}} = \rho_L L_{\text{bulk}} + \rho_{LM} M_{\text{bulk}}.$$

Substitution in [FH, eqs. (3.32), (3.34)] gives eight independent polynomial equations after including the central branch and crossed bracket. Their only solutions are the two globally conjugate branches. The selected branch is

$$\boxed{A = B = \frac{ia_2}{2}, \quad C = 0, \quad D = i, \quad \rho_M = \rho_L = 2a_2^2, \quad \rho_{LM} = 0.}$$

The same sign for A and B is forced by the FH factor i. Indeed,

$$[J_m^+, J_n^-]_{\text{cl}} = -AB\,[\psi'_m, \bar{\psi}'_n]_{\text{cl}} = \frac{m}{4}\delta_{m+n,0}.$$

The extra minus sign is $(-i)^2$, because the ordinary Fourier coefficient of J_{FH} is iJ_n. Reversing only the barred current gives $-m/4$ and fails.

Equivalently, in terms of the diagonal matrices,

$$\boxed{\mathcal{P}_j = -a_s(D_0)_{jj}, \quad \mathcal{N}_j = -a_s(D_1)_{jj}, \quad P_j^{\text{FH}} = -ia_s(D_0)_{jj}, \quad N_j^{\text{FH}} = -ia_s(D_1)_{jj}.}$$

```
import sympy as s
A,B,C,D,rM,rL,rLM,a=s.symbols('A B C D rM rL rLM a')
eq=[2*A**2+rM/4,2*D*a*A+rM/2,4*A*B+rL/2,2*D*a*B+rL/2,
    4*A*C+rLM/4,2*D*a*C+rLM/2,D**2+1,A*B+a**2/4]
sol=s.solve(eq,[A,B,C,D,rM,rL,rLM],dict=True)
target={A:s.I*a/2,B:s.I*a/2,C:0,D:s.I,rM:2*a**2,rL:2*a**2,rLM:0}
assert len(sol)==2 and target in sol
print('CELL_IV_3_PASS')
```

6. N=2 screening charges and generated transformations

The primitive map follows by differentiating the current map and fixing the inherited basepoint constants:

$$\boxed{\mu\Phi^+ = -\psi, \qquad \mu\Phi^- = -\bar{\psi}.}$$

Therefore

$$E^{\text{cl}} = e^{-\psi}, \qquad \mathcal{S}_{\text{odd}}^{[1],\text{cl}} = -\bar{\psi}e^{-\psi}.$$

These are precisely the two integrands of [DDF, eq. (4.35)]:

$$S_{\kappa_4} = \frac{\kappa_4}{4\pi G}\int e^{-\psi}\,d\phi, \qquad S_{\kappa_2} = -\frac{\kappa_2}{4\pi G}\int \bar{\psi}e^{-\psi}\,d\phi.$$

The bracket gives [DDF, eq. (4.40)]

$$\{S, \psi'\} = 2\kappa_2 e^{-\psi}, \qquad \{S, \bar{\psi}'\} = 2(\kappa_4 - \kappa_2\bar{\psi})e^{-\psi}.$$

Substitution shows $\delta M_{\text{bulk}} = \delta L_{\text{bulk}} = 0$ coefficientwise. If the boundary circle charges are normalized as unit integrals, then

$$Q^{[0],\text{cl}} = \frac{4\pi G}{\kappa_4}S_{\kappa_4}, \qquad \widetilde{Q}^{[1],\text{cl}} = \frac{4\pi G}{\kappa_2}S_{\kappa_2}.$$

7. The general diagonal bracket has a factor-two inconsistency

Equation [DDF, (6.23c)] prints

$$\{(D_0)_{ii}, (D_1)_{jj}\}_{\text{print}} = -\frac{1}{2k_{\text{CS}}\tau_s}\left(\delta_{ij} - \frac{1}{s}\right)\delta'.$$

This is incompatible with both explicit sectors after using the paper's own diagonal map (6.49).

For $s = 2$, its $i = j = 1$ component is $-\pi G\delta'$, whereas (4.39) and $D_{11} = \psi'/2$ give $-2\pi G\delta'$. For $s = 3$, the printed component is $-16\pi G\delta'/3$, whereas (5.23) and

$$(D_0)_{11} = \psi'_{(L)} + \frac{2}{3}\psi'_{(W)}$$

give $-32\pi G\delta'/3$. Both explicit sectors agree with

$$\boxed{\{(D_0)_{ii}, (D_1)_{jj}\}_{\text{explicit}} = -\frac{1}{k_{\text{CS}}\tau_s}\left(\delta_{ij} - \frac{1}{s}\right)\delta'.}$$

Machine token: `1/(kCS*tau_s)`. Notebook IV uses the explicit $s = 2, 3$ brackets for the theorem and records the disagreement rather than absorbing it into separate field rescalings.

8. A2 weights, diagonal map, and root symplectic form

Choose the fundamental weights

$$\epsilon_1 = \left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{6}}\right), \quad \epsilon_2 = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{6}}\right), \quad \epsilon_3 = \left(0, -\frac{2}{\sqrt{6}}\right).$$

They obey

$$\epsilon_i \cdot \epsilon_j = \delta_{ij} - \frac{1}{3}, \qquad \sum_i \epsilon_i = 0, \qquad \sum_i \epsilon_i \epsilon_i^{\mathrm{T}} = I_2.$$

With $J_0 = \text{diag}(1, 0, -1)$ and $U_0 = \frac{2}{3}\text{diag}(1, -2, 1)$, the explicit diagonal field [DDF, eq. (5.12)] is

$$D_0 = \text{diag}\left(\psi'_{(L)} + \frac{2}{3}\psi'_{(W)}, -\frac{4}{3}\psi'_{(W)}, -\psi'_{(L)} + \frac{2}{3}\psi'_{(W)}\right),$$

and similarly for D_1. Equation (6.49) fixes one map in both sectors,

$$\psi = -ia_3 \begin{pmatrix} 1/\sqrt{2} & \sqrt{2} \\ \sqrt{6}/2 & -\sqrt{6}/3 \end{pmatrix} \begin{pmatrix} \psi_{(L)} \\ \psi_{(W)} \end{pmatrix},$$

with the same matrix for $\bar{\psi}$.

Define $\alpha_1 = \epsilon_1 - \epsilon_2$, $\alpha_2 = \epsilon_2 - \epsilon_3$. Then

$$\alpha_i \cdot \alpha_j = A_{ij}, \qquad A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix},$$

and

$$\chi_1 = \psi_{(L)} + 2\psi_{(W)}, \qquad \chi_2 = \psi_{(L)} - 2\psi_{(W)}$$

satisfy

$$\chi_i = \frac{i}{a_3}\alpha_i \cdot \psi.$$

The explicit brackets [DDF, eq. (5.23)] give

$$\boxed{\{\chi_i'(\phi),\bar{\chi}_j(\phi')\}=16\pi G A_{ij}\delta(\phi-\phi').}$$

```
import sympy as s
R=s.Matrix([[1,2],[1,-2]])
A=s.Matrix([[2,-1],[-1,2]])
assert R*s.diag(8,6)*R.T==16*A
E=s.Matrix([[1/s.sqrt(2),1/s.sqrt(6)],[-1/s.sqrt(2),1/s.sqrt(6)],[0,-2/s.sqrt(6)]])
assert s.simplify(E*E.T-(s.eye(3)-s.ones(3)/3))==s.zeros(3)
print('CELL_IV_4_PASS')
```

9. Four A2 Miura observables

The primitive dictionary is now forced, rather than guessed:

$$\boxed{\mu\Phi_i^+=-\chi_i,\qquad \mu\Phi_i^-=-\bar{\chi}_i.}$$

Machine tokens: `mu*Phi_i_plus=-chi_i` and `mu*Phi_i_minus=-bar_chi_i`.

Differentiating and solving the traceless constraints gives

$$P_1=-\frac{2\chi_1'+\chi_2'}{3\mu},\qquad P_2=\frac{\chi_1'-\chi_2'}{3\mu},$$

$$N_1=-\frac{2\bar{\chi}_1'+\bar{\chi}_2'}{3\mu},\qquad N_2=\frac{\bar{\chi}_1'-\bar{\chi}_2'}{3\mu}.$$

Substitution into the frozen primary Miura fields yields four exact identities:

$$\boxed{U_2^+=-\frac{4}{\mu^2}M_{(L)},\qquad U_2^-=-\frac{4}{\mu^2}L_{(L)},}$$

$$\boxed{\mathcal{U}_3^+=-\frac{8}{\mu^3}M_{(W)},\qquad \mathcal{U}_3^-=-\frac{8}{\mu^3}L_{(W)}.}$$

No root-dependent factor and no spin-mixing improvement is present.

9.1 The printed spin-three polynomial is inconsistent

Equation [DDF, (5.28c)] prints a term $9(\psi''_{(L)})^2$ inside the parenthesis multiplying $\psi'_{(W)}$. Its differential weight is five, although $M_{(W)}$ has spin three. The general row determinant (6.51) and the coefficient solve both require a linear $9\psi''_{(L)}$. Thus the expression used above contains

`9*psi_L_double_prime`

and not its square. Reinstating the square makes the $\mathcal{U}_3^+$ identity fail.

```
import sympy as s
mu=s.symbols('mu',nonzero=True)
x1,x2,x1p,x2p,x1pp,x2pp=s.symbols('x1 x2 x1p x2p x1pp x2pp')
P1=-(2*x1+x2)/(3*mu);P2=(x1-x2)/(3*mu);a=1/mu
dP1=-(2*x1p+x2p)/(3*mu);dP2=(x1p-x2p)/(3*mu);d2P2=(x1pp-x2pp)/(3*mu)
U3=s.expand(-P1**2*P2-P1*P2**2+a*(s.Rational(3,2)*P1*dP2+s.Rational(1,2)*dP1*P2+P2*dP2)-a**2*d2P2/2)
l=(x1+x2)/2;w=(x1-x2)/4;lp=(x1p+x2p)/2;wp=(x1p-x2p)/4;wpp=(x1pp-x2pp)/4
MW=s.expand((w*(18*l**2-8*w**2+9*lp)+27*l*wp+9*wpp)/108)
assert s.simplify(U3+8*MW/mu**3)==0
print('CELL_IV_5_PASS')
```

10. A2 simple-root screenings and generated transformations

The four simple charges [DDF, eq. (5.26)] are

$$S_{8,i} = \frac{\kappa_{8,i}}{8\pi G}\int e^{-\chi_i}\,d\phi, \qquad S_{2,i} = -\frac{\kappa_{2,i}}{8\pi G}\int \bar{\chi}_i e^{-\chi_i}\,d\phi,$$

where $(\kappa_{8,1},\kappa_{8,2}) = (\kappa_8,\kappa_9)$ and $(\kappa_{2,1},\kappa_{2,2}) = (\kappa_2,\kappa_3)$. The exponent labels fix

$$Q_1^{[0]} \leftrightarrow S_{\kappa_8}, \qquad Q_2^{[0]} \leftrightarrow S_{\kappa_9}.$$

The root bracket gives the transformations

$$\delta_{8,i}\chi_j' = 0, \qquad \delta_{8,i}\bar{\chi}_j' = 2\kappa_{8,i}A_{ij}e^{-\chi_i},$$

$$\delta_{2,i}\chi_j' = 2\kappa_{2,i}A_{ij}e^{-\chi_i}, \qquad \delta_{2,i}\bar{\chi}_j' = -2\kappa_{2,i}A_{ij}\bar{\chi}_i e^{-\chi_i}.$$

Applying these variations, with $D(e^{-\chi_i}) = -\chi_i' e^{-\chi_i}$, annihilates all four observables for both roots. The main verifier performs all sixteen root/charge/observable checks.

For unit circle-integral normalization,

$$Q_i^{[0],\mathrm{cl}} = \frac{8\pi G}{\kappa_{8,i}}S_{8,i}, \qquad \widetilde{Q}_i^{[1],\mathrm{cl}} = \frac{8\pi G}{\kappa_{2,i}}S_{2,i}.$$

```
import sympy as s
G=s.symbols('G',positive=True);pi=s.pi
A=s.Matrix([[2,-1],[-1,2]])
assert s.simplify((1/(8*pi*G))*(16*pi*G)-2)==0
assert s.simplify((-1/(8*pi*G))*(-16*pi*G)-2)==0
chi,bchi=s.symbols('chi bchi')
assert s.exp(-chi)==s.exp(-chi)
assert -bchi*s.exp(-chi)==-bchi*s.exp(-chi)
print('CELL_IV_6_PASS')
```

11. Even/affine coefficient and BCH power counting

At the gravity point, $\alpha^- = \nu = 0$. Gate 3B gives

$$\mathcal{S}_{i,\mathrm{odd}}^{[1]} =: \mu\Phi_i^- E_i :,$$

and

$$\mathcal{S}_{i,\mathrm{even}}^{[1]} =: \left(-\mu^3\Phi_i^+ + \mu^2\log(w/w_0)\right)E_i : .$$

Since $\mu\Phi_i^+ = O(1)$ and $\mu^2 = O(G)$,

$$\mathcal{S}_{i,\mathrm{odd}}^{[1]} = O(1), \qquad \mathcal{S}_{i,\mathrm{even}}^{[1]} = O(G).$$

The barred BCH conversion is

$$\bar{Z}_{t,\rho} = (1-\rho)^{-2t\bar{\mu}(t)^2}, \qquad \log\bar{Z}_{t,\rho} = -2t\bar{\mu}(t)^2\log(1-\rho) = O(tG).$$

It is scalar and insertion-coordinate independent. It changes only the allowed triangular multiple of $Q_i^{[0]}$; it cannot supply a new $O(G^0)$ local charge.

```
import sympy as s
G,c,t,rho=s.symbols('G c t rho')
mu2=c*G
logZ=-2*t*mu2*s.log(1-rho)
```

```
assert s.diff(logZ,t).has(G)
qraw,q0,qm,mu=s.symbols('qraw q0 qm mu')
qcanonical=qraw-mu*qm*q0
assert s.expand(qcanonical-(qraw-mu*qm*q0))==0
print('CELL_IV_7_PASS')
```

12. Raw moving frame and canonical PBW frame

The bulk Noether charges act on one classical phase space and annihilate the Miura invariants homogeneously. They therefore realize the raw horizontal coefficient:

$$\boxed{S_i^{(1)} \longleftrightarrow \widetilde{Q}_i^{[1],\mathrm{cl}}.}$$

At $\nu = 0$, decompose

$$\Phi_i^- = q_i^- + p_i^- \log z + \Phi_{i,\mathrm{osc}}^-.$$

The bulk field $\bar{\chi}_i$ includes its constant mode unless a basepoint condition is imposed. Hence the bulk barred-linear charge contains the image of $\mu q_i^- Q_i^{[0]}$ and is the raw representative. A fixed-target PBW trivialization subtracts precisely this target-frame derivative:

$$\boxed{Q_i^{[1]} = \widetilde{Q}_i^{[1]} - \mu\, \alpha_i \cdot q^-\, Q_i^{[0]}.}$$

Changing the bulk basepoint shifts $\bar{\chi}_i$ by a constant and therefore changes the barred-linear charge by a scalar multiple of the leading charge. This is the same triangular ambiguity already allowed by Gate 3B; it does not create an additional screening.

13. Composite screening diagnostic

The nonlocal composite charges in [DDF, eq. (5.27)] depend on a basepoint ϕ_0. The source also notes below eq. (6.37) that higher-upper-diagonal transformations can involve field-dependent integration constants. Notebook IV therefore assigns the optional composite gate Outcome C: no basepoint-independent quantum screening-screening conclusion is extracted. This does not affect the simple-root theorem.

14. Theorems and outcome

Theorem IV.1 - exact semiclassical normalization

Let

$$k_{\mathrm{CS}} = \frac{1}{8\pi G}, \qquad \tau_s = \frac{12}{s(s^2-1)}, \qquad a_s = \sqrt{\pi k_{\mathrm{CS}} \tau_s},$$

and choose $\alpha^- = 0$, $\alpha^+ = i a_s$. Then $c_L = 0$, $c_M = 3/G$. For $s = 2, 3$, the single scalar-to-weight map, the FH factor i, and the common current map derived above identify the bulk row-determinant invariants with the semiclassical symmetric Carrollian Miura fields. For $s = 3$, the exact relations are

$$U_2^{\pm} = -\frac{4}{\mu^2}(M_{(L)}, L_{(L)})^{\pm}, \qquad \mathcal{U}_3^{\pm} = -\frac{8}{\mu^3}(M_{(W)}, L_{(W)})^{\pm}.$$

Theorem IV.2 - simple-root screening map

For $s = 2, 3$,

$$E_i^{\mathrm{cl}} = e^{-\chi_i}, \qquad \mathcal{S}_{i,\mathrm{odd}}^{[1],\mathrm{cl}} = -\bar{\chi}_i e^{-\chi_i}.$$

After fixing the bulk gauge parameters, $Q_i^{[0],\mathrm{cl}}$ and $\widetilde{Q}_i^{[1],\mathrm{cl}}$ equal the corresponding leading and barred-linear bulk Noether charges up to the explicitly displayed universal prefactors. Both root labels and both absolute signs are fixed by generated transformations. The common even/affine coefficient is $O(G)$.

Corollary IV.3 - canonical-frame interpretation

The classical bulk barred-linear screening realizes the raw horizontal coefficient. The canonical $Q_i^{[1]}$ is its PBW-covariantized deformation intertwiner. It is not an ordinary second quantum screening.

Disposition

Notebook IV closes with Outcome B.

The only convention transformation is one global analytic continuation and plane/cylinder conversion, used uniformly in the $s = 2$, $s = 3$, unbarred, barred, current, Miura, and screening sectors. There is no root-dependent normalization.

15. Claim boundary

This notebook does not establish a quantum screening doublet, equality of quantum kernels, a BRST complex, a quantum-group structure, quantum Serre relations, complete screening-screening closure, equality of quantum correlation functions, or a full holographic duality. The two primary-source formula inconsistencies and the general-bracket factor-two inconsistency are local source audits, not claims that the remaining results of either paper fail.

16. Reproducibility ledger

Layer	Required evidence
Upstream	five exact archive hashes; portable Gate-3B checkpoint
Main model	normalization, Fourier, $s = 2$, A_2, four Miura fields, transformations, power counting, frame
Independent model	no import from the main verifier; separate 29-check reconstruction
Source anchors	dedicated $s = 2$ and $s = 3$ programs
Embedded cells	seven isolated SymPy processes
Adversarial	all roadmap mutations plus real unmutated controls
Archive	strict manifest, safe paths, exact payload set, fresh extraction
Static record	A4, embedded fonts, page-by-page rendered inspection, HTML/PDF theorem synchronization

Synchronization tokens: `raw horizontal coefficient`; `i/(2*sqrt(G))`; `9*psi_L_double_prime`; `1/(kCS*tau_s)`; `mu*Phi_i_minus=-bar_chi_i`; `not an ordinary second quantum screening`; `Outcome B`; `NIV-36a2530aab4feb7d`.

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